



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics and Telecommunication Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (Semester I)

Class: TY B.Tech

Course: Control Systems

Course Code: PCC-EC302/PCC-EC303

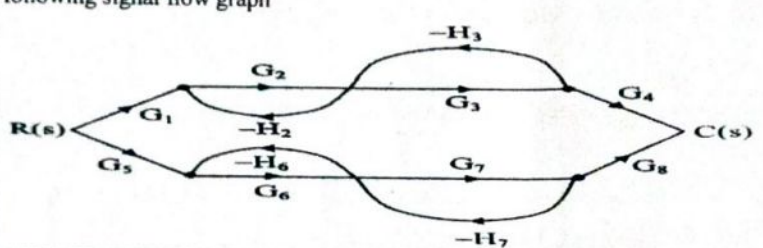
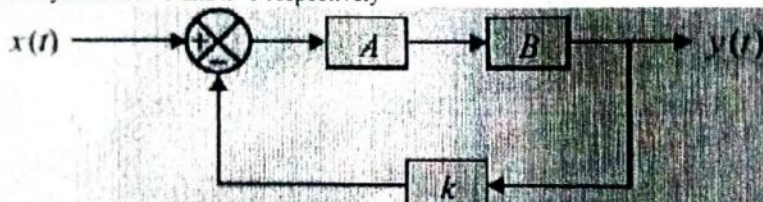
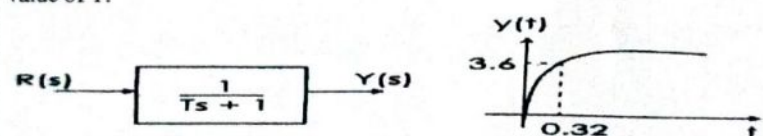
Date: 14.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	(a) Name the test signals used in time response analysis of a control system. (b) Why are ramp and parabolic input functions not used for the analysis of a first-order control system? Explain with proper reasoning and examples.	07	CO2	BT1, BT2, BT3, BT4
Q.2	Using Mason's Gain formula determine the overall transfer function for the following signal flow graph 	07	CO1	BT1, BT2, BT3
Q.3	The figure shows the block diagram representation of a control system. The system in block A has an impulse response $h_A(t) = e^{-t} u(t)$. The system in block B has an impulse response $h_B(t) = e^{-2t} u(t)$. Determine the impulse response of the system at $k=1$ and $k=0$ respectively 	04	CO3	BT1, BT2
Q.4	A system and its response to an input $r(t) = 8u(t)$ is given below. Determine the value of T. 	02	CO3	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO1	Model a physical system and express its internal dynamics and input-output relationships by means of block diagrams, mathematical model and transfer functions.
CO2	Understand and explain the relationships between the parameters of a control system and its stability, accuracy, transient behaviour.
CO3	Identify the parameters that the system is sensitive to. Determine the stability of a system and parameter ranges for a desired degree of stability.
CO4	Plot the Bode, Nyquist, Root Locus diagrams for a given control system and identify the parameters and carry out the stability analysis and to determine the frequency response of a control system and use it to evaluate or adjust the relative stability.
CO5	Design a P, PD, PI, or PID controller based on the transient and steady state response criteria and Model and analyse the control systems using state space analysis.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE II Examination (Semester-I) 2025-26

T.Y. B. Tech.

Electronics & Telecommunication Engineering

Electromagnetic Engineering

PCC-EC301

Date: 15/10/2025

Time: - 11:00 to 11:45

Max. Marks: 20

- Note:
- Attempt all questions
 - Assume suitable data if necessary
 - Use of non-programmable calculator is permitted.

Q.N.	Questions	Marks	CO	BT
Q.1	Explain and Derive boundary conditions for electric field intensity for a dielectric- dielectric interface.	[5]	CO4	BT2
Q.2	A potential field is given as $V=100e^{-5x} \sin 3y \cos 4z$ V. Let point P $(0.1, \pi/12, \pi/24)$ be located at a conductor free space boundary. At point, Find the magnitudes of a) V b) $\vec{E}$ c) E_t d) E_N e) $\vec{D}$.	[5]	CO4	BT3
Q.3	If the three charges $3\mu\text{C}$, $4\mu\text{C}$ and $5\mu\text{C}$ are located at $(0,0,0)$, $(2,1, -3)$ and $(0,4, -2)$ respectively. Find the potential at $(1,0,1)$ assuming zero potential at infinity.	[5]	CO3	BT4
Q.4	Explain and Derive boundary conditions for $\vec{E}$ between conductor and free space.	[5]	CO4	BT 2

Course Outcomes (CO)

	On successful completion of this course, students should be able to:
CO1	Understanding the concepts of vector fields E, D, H & B.
CO2	Cleared the Concepts Divergence and Stokes theorem.
CO3	Get an idea of the concepts: Energy, Work done, Potential, Potential gradient and dipole
CO4	Get the idea of the terms Conductors, Dielectrics, boundary conditions and capacitance
CO5	Understanding of Poisson's and Laplace's equations.
CO6	Get knowledge about Time Varying Field and Maxwell's Equations
CO7	Get an idea of Uniform Plane Wave used for propagation

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

**SGGS Institute of Engineering and Technology, Vishnupuri, Nanded. ISE-1 Examination,
Academic Year 2025-26 (Sem1)**

Class: Third Year (Electronics and Telecommunication Engineering)

Date: 15/10/25

Subject: Microcontrollers and Applications,

Time: 11 to 11:45 p.m.

Subject code: PCC-EC 301

Max. Marks : 20 , Duration : 45 minutes

Note: I. Use of calculator is not allowed

II. All questions are compulsory to solve.

Sr. No.	Questions	CO	BT	Marks
Q.1	Explain TMOD and TCON Registers.	CO3	BT1 BT2	05
Q.2	What value is to be loaded in TL0 and TH0 register to get largest delay possible? What is the largest delay? Use 11.0592MHZ crystal. Find the delay generated by timer 0 in the following code. <pre> CLR P2.3 ; MOV TMOD, #01 ; HERE: MOV TL0, #11H ; MOV TH0, #0FFH ; SETB P2.3 ; SETB TR0 ; AGAIN: JNB TF0, AGAIN ; CLR TR0 ; CLR TF0 ; CLR P2.3 ; SJMP HERE </pre>	CO2 CO3	BT3	05
Q.3	Write a program to blink all the LEDS connected to port P1 so that the blinking is clearly seen. Use 8051 crystal frequency as per your choice. Create a square wave of 66% duty cycle on bit 2 of port 1 using bit manipulation instructions.	CO2 CO3	BT3 BT4	05
Q.4	Differentiate between Serial and parallel type of communication. What do you mean by Full Duplex data transfer scheme.	CO4	BT1 BT2	05
	END			
COs	CO1: Interpret architecture of 8-bit microprocessor and microcontrollers CO2: Develop program in 8051 in assembly language for the given operation. CO3: Develop program using timers and interrupts. CO4: Interface the memory and I/O peripherals to 8051 microcontrollers. CO5: Design microcontroller-based applications.			
BT	BT1: Remember, BT2: Understand, BT3: Apply, BT4: Analyze, BT1: Evaluate, BT2: Create			



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering
In-Semester Examination II
Academic Year: 2024-25 (Semester – I)

Class: TY B. Tech	Subject: Statistical Machine Learning	Subject Code: MDM IS – EC301
Date: 16.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.

Q1	State and explain Minimum Error Rate classification.	5 M	CO 1
Q2	State and explain Bayes Decision Theorem for single feature.	5 M	CO 1
Q3	Feature x is uniformly distributed in the range $0 \leq x \leq 2$ for class A and uniformly distributed in the range $1 \leq x \leq 4$ for class B. The prior probabilities are $P(A) = 0.6$ and $P(B) = 0.4$. Find $P(A/x)$ when $x = 1.5$.	10 M	CO 2

== ALL THE BEST ==

Course Outcomes:

CO 1	Describe concepts of machine learning and it's learning and adaptation for the Classification.
CO 2	Understand the principles of Bayesian and Maximum likelihood parameter estimation and apply them in relatively simple probabilistic models such as Gaussian Model.
CO 3	Perform data clustering in an unsupervised manner by means of various algorithms, such as Parzen window density Estimate.
CO 4	Understand the concept of classification using K-means and K-nearest neighbour and it's Applications.
CO 5	Learn un-supervised learning techniques.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics & Telecommunication Engg.
In-Semester Examination - II
Academic Year: 2025-26 (semester -I)

Class: TY

Course: Digital system Design
using Verilog (Elective)

Course Code: PEC-EC305/PEC-EC313

Date: 09.10.2025

Time: 11.00 PM to 11:45 PM

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		<u>Solve Any Two Question</u>	Marks	CO	BT
Q.1	a	Design a Parameterized Arithmetic Logic Unit(ALU) Write Verilog code for it.	5	CO3	BT6
	b	Explain the concept of Two's complement binary numbers with an example	5	CO2	BT2
Q.2	a	Design a 2:4 Decoder circuit modules in Verilog.	5	CO3	BT6
	b	Write Verilog code for the encoder using any one of them: (i) Dataflow modelling (ii) Behavioral modelling	5	CO2	BT3
Q.3	a	Design a Comparator using Verilog. Write code to compare two 1bit numbers and display whether $A > B$, $A = B$, or $A < B$.	5	CO3	BT6
	b	Design a 2-to-1 Multiplexer using Verilog. Show how this can be extended to build an 8-to-1 Multiplexer .	5	CO3	BT6

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understanding Digital Design trends and software design aspects.
CO 2	Understanding Combinational and Sequential design aspects.
CO 3	Implementation of Combinational and Sequential Circuits using Verilog HDL.
CO 4	Understanding Data path and Control Path design aspects.
CO 5	Implementation of Data path and Control Path using Verilog HDL.



T.Y. B. Tech.
Analog Communication
Date: 11/10/2025 Time: 11.00 – 11.45

Electronics and Telecommunication Engineering
PCC-EC305
Max. Marks: 20

- Note:
- Attempt all questions
 - Assume suitable data if necessary.
 - Use of a non-programmable calculator is permitted.
 - This paper contains 01 page.

Q.N.	Question	Marks	CO	BT
Q.1	A signal $m_1(t)$ is band limited to 4.8 kHz, and three other signals $m_2(t)$, $m_3(t)$ and $m_4(t)$ are band limited to 1.2 kHz each. These signals are to be transmitted by means of time division multiplexing. - Set up a scheme for accomplishing this multiplexing requirement, with each signal sampled at its Nyquist rate. - What must be the speed of the commutator (in samples per second)? - If the commutator output is quantized with $L=4096$ and the result is binary coded, what is the output bit rate? - Determine the minimum transmission bandwidth of the channel.	[06]	PCC-EC-303.1	BT2
Q.2	A compact disc (CD) recording system samples each of three stereo signals with a 32-bit analog-to-digital converter (ADC) at 40 kb/s. - Determine the output signal-to-quantizing-noise ratio for full-scale sinusoid. - The bit stream of digitized data is augmented by the addition of error-correcting bits, clock extraction bits and control bit fields. These additional bits represent 100% overhead. Determine the output bit rate of the CD recording system. - The CD can record an hour's worth of music. Determine the number of bits recorded on a CD. - For a compression, a high-grade collegiate may contain 1500 pages, 2 columns per page, 100 lines per column, 8 words per line, 6 letters per word and 7 bits per letter on average. Determine the number of bits required to describe the dictionary and estimate the number of comparable books that can be stored on a CD.	[08]	PCC-EC-303.3	BT3
Q.3	A PCM system uses a uniform quantizer followed by a 8-bit binary encoder. The bit rate of the system is equal to 50 Mbps. - What is the maximum message bandwidth for which the system operates satisfactorily? - Determine the output signal-to-quantization noise ratio when a full loaded sinusoidal modulating wave of frequency 1 MHz is applied to the input.	[06]	PCC-EC-303.2	BT1

Course Outcomes (CO)

PCC-EC-303.1- Understand the mathematical modelling of noise.

PCC-EC-303.2- Analyze the behavior of analog and digital modulation systems in presence of noise.

PCC-EC-303.3- Analyze the effect of channel noise on digital communication system.

PCC-EC-303.4- Design the optimum filter like Matched filter to optimize the detector performance.

PCC-EC-303.5- Understand the effect of inter symbol interference (ISI) in channel modelling.

Bloom's Taxonomy (BT) BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
ISE-II Examination AY 2025-26

Class : SY/TY(EXTC)

Date: 13/10/2025

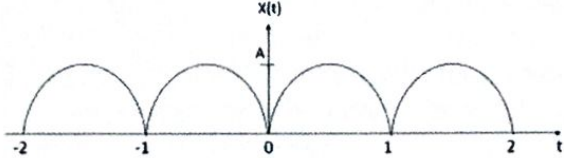
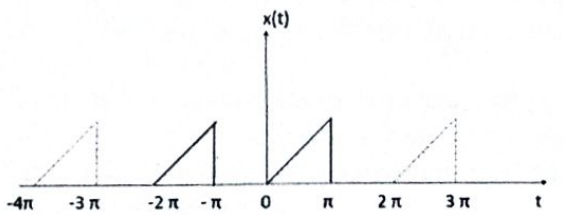
Time: 11:00-11:45

Exam duration : 45 min

Subject: Signals and Systems

Course Code: PCC-EC202/EC-304

Max Marks: 20

Q.No.	Questions	Marks	COs	BT
Q. 1	Obtain the exponential Fourier series of the waveform shown in figure. Also draw frequency spectrum 	6	CO1, CO2	BT4
Q2	Obtain the trigonometric Fourier series for the waveform shown in Figure. 	6	CO1, CO2	BT3
Q3	State & derive the properties of the Fourier series.(any 2) 1. Time differentiation Property 2. Convolution Property 3. Time Reversal Property	5	CO2	BT1
Q5	Find the Fourier transform of the given function. State the importance of the Fourier transform. $X(t) = \cos(\omega_0 t)$	3	CO1, CO2	BT1

Blooms' Taxonomy:

BT1: REMEMBER, BT2: UNDERSTAND, BT3: APPLY, BT4: ANALYSIS, BT5: EVALUATE, BT6: CREATE

Course Outcome:

At the end of Course Students will be able.

CO1: Analyse different types of signals.

CO2: Represent continuous and discrete systems in time and frequency domain using Different transforms.

CO3: Investigate whether the system is stable.

CO4: Sampling and reconstruction of a signal.

CO5: Classify between different types of systems and signals.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE-II Examination

AY 2025-26 Sem-I

T.Y. B. Tech.

Communication Systems Theory

Electronics and Telecommunication Engineering

PCC-EC303

Max. Marks: 20

Date: 11/10/2025

Time: 11.00 – 11.45

- Note:
- Attempt all questions
 - Assume suitable data if necessary.
 - Use of a non-programmable calculator is permitted.
 - This paper contains 01 page.

Q.N.	Question	Marks	CO	BT
Q.1	A signal $m_1(t)$ is band limited to 4.8 kHz, and three other signals $m_2(t)$, $m_3(t)$ and $m_4(t)$ are band limited to 1.2 kHz each. These signals are to be transmitted by means of time division multiplexing. - Set up a scheme for accomplishing this multiplexing requirement, with each signal sampled at its Nyquist rate. - What must be the speed of the commutator (in samples per second)? - If the commutator output is quantized with $L=4096$ and the result is binary coded, what is the output bit rate? - Determine the minimum transmission bandwidth of the channel.	[06]	PCC-EC-303.1	BT2
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Q.3	A PCM system uses a uniform quantizer followed by a 8-bit binary encoder. The bit rate of the system is equal to 50 Mbps. - What is the maximum message bandwidth for which the system operates satisfactorily? - Determine the output signal-to-quantization noise ratio when a full loaded sinusoidal modulating wave of frequency 1 MHz is applied to the input.	[06]	PCC-EC-303.2	BT1

Course Outcomes (CO)

PCC-EC-303.1- Understand the mathematical modelling of noise.

PCC-EC-303.2- Analyze the behavior of analog and digital modulation systems in presence of noise.

PCC-EC-303.3- Analyze the effect of channel noise on digital communication system.

PCC-EC-303.4- Design the optimum filter like Matched filter to optimize the detector performance.

PCC-EC-303.5- Understand the effect of inter symbol interference (ISI) in channel modelling.

Bloom's Taxonomy (BT) BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE-II Examination (Semester-I) 2025-26

Third Year B. Tech.

Electronics and Telecommunication
Engineering

Probability and Statistics - P&S

PEC - EC301

Date: 10/
Oct/2025 (FRI)

Time: 11:00 to 11:45

Max. Marks : 20

Note: i. Assume suitable data if necessary. ii. Use of non-programmable calculators is permitted.

iii. This paper contains one page only. iv. **Solve all questions.**

Q.N	Question	Marks	CO	BT
Q.1	The probability density function (pdf) of a continuous random variable X is proportional to x^2 on the interval $0 < x \leq 8$. If the median of the distribution is 6.34, find the mode of X .	05	1, 3	3, 4
Q.2	State and prove Bayes Theorem with a neat sketch. What is Central Limit Theorem?	05	1, 2	2, 4
Q.3	Differentiate between Measures of central tendency and Measures of spread.	05	4	2
Q.4	Explain the following Distributions. i) Binomial distribution ii) Uniform distribution iii) Normal distribution	05	3	1

*****!!!!!!All the Best!!!!!!*****

Details below are for Information only:

Bloom's Taxonomy: BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create
COs: Course Outcomes

CO1: Understand the basic knowledge on fundamental probability concepts, including random variable, probability of an event, additive rules and conditional probability

CO2: Apply the central limit theorem to sampling distribution and use estimation technique to determine point estimates confidence interval and sample size.

CO3: Understand several well-known distributions, including Binomial, Geometrical, Negative Binomial, Normal and Exponential Distribution.

CO4: Perform and analyse hypotheses tests of means, proportions and variances using both one- and two-sample data sets.

CO5: Compute and interpret all of the results of linear Regression.